



Institut Universitaire des sciences

- Faculté des sciences et de technologie
-
- ✓ TD N°4 Administration Système

Nom & Prénom : COFFY Cliford

Niveau : L3

Prof. : Ismael SAINT AMOUR

Date : 23/03/2026

1. Ici je commence à reproduire les taches

On commence avec la commande : « ss -tuln »

- ✓ On fait cela pour identifier les ports ouverts

```
clif@ubuntuserver: ~$ ss -tuln
Netid State  Recv-Q Send-Q           Local Address:
Port                               Process
udp    UNCONN  0      0      0.0.0.0:*      127.0.0.54:
53
udp    UNCONN  0      0      0.0.0.0:*      127.0.0.53%lo:
53
udp    UNCONN  0      0      0.0.0.0:*      192.168.1.115%enp0s3:
68
udp    UNCONN  0      0      0.0.0.0:*      0.0.0.0:
111
udp    UNCONN  0      0      0.0.0.0:*      127.0.0.1:
161
udp    UNCONN  0      0      0.0.0.0:*      [::]:
111      [::]:*
udp    UNCONN  0      0      [::]:*      [:::1]:
161      [::]:*
udp    UNCONN  0      0      [fe80::a00:27ff:fe97:c322]%enp0s3:
546      [::]:*
tcp    LISTEN  0      4096      0.0.0.0:
111      0.0.0.0:*
tcp    LISTEN  0      100      0.0.0.0:
25      0.0.0.0:*
tcp    LISTEN  0      4096      0.0.0.0:
22      0.0.0.0:*
tcp    LISTEN  0      4096      127.0.0.53%lo:
53      0.0.0.0:*
tcp    LISTEN  0      4096      127.0.0.54:
53      0.0.0.0:*
tcp    LISTEN  0      4096      0.0.0.0:
10050   0.0.0.0:*
tcp    LISTEN  0      4096      [::]:
```

```
clif@ubuntuuserver: ~  
111 0.0.0.0:*  
udp UNCONN 0 0 127.0.0.1:  
161 0.0.0.0:*  
udp UNCONN 0 0 [::]:  
111 [::]:*  
udp UNCONN 0 0 [::1]:  
161 [::]:*  
udp UNCONN 0 0 [fe80::a00:27ff:fe97:c322]%enp0s3:  
546 [::]:*  
tcp LISTEN 0 4096 0.0.0.0:  
111 0.0.0.0:*  
tcp LISTEN 0 100 0.0.0.0:  
25 0.0.0.0:*  
tcp LISTEN 0 4096 0.0.0.0:  
22 0.0.0.0:*  
tcp LISTEN 0 4096 127.0.0.53%lo:  
53 0.0.0.0:*  
tcp LISTEN 0 4096 127.0.0.54:  
53 0.0.0.0:*  
tcp LISTEN 0 4096 0.0.0.0:  
10050 0.0.0.0:*  
tcp LISTEN 0 4096 [::]:  
111 [::]:*  
tcp LISTEN 0 511 *: *:   
80 *: *:   
tcp LISTEN 0 100 [::]:  
25 [::]:*  
tcp LISTEN 0 4096 [::]:  
22 [::]:*  
tcp LISTEN 0 4096 [::]:  
10050 [::]:*  
clif@ubuntuuserver:~$ |
```

Quand on fait avec « netstat -tuln »

```
clif@ubuntuuserver: ~  
clif@ubuntuuserver:~$ netstat -tuln  
Active Internet connections (only servers)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:111             0.0.0.0:*               LISTEN  
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN  
tcp        0      0 0.0.0.0:10050            0.0.0.0:*               LISTEN  
tcp6       0      0 :::111                  :::*                    LISTEN  
tcp6       0      0 :::80                   :::*                    LISTEN  
tcp6       0      0 :::25                   :::*                    LISTEN  
tcp6       0      0 :::22                   :::*                    LISTEN  
tcp6       0      0 :::10050                 :::*                    LISTEN  
udp        0      0 127.0.0.54:53           0.0.0.0:*                 
udp        0      0 127.0.0.53:53           0.0.0.0:*                 
udp        0      0 192.168.1.115:68        0.0.0.0:*               
```

```
clif@ubuntuuserver: ~  
LISTEN  
tcp 0 0 127.0.0.54:53 0.0.0.0:*  
LISTEN  
tcp 0 0 0.0.0.0:10050 0.0.0.0:*  
LISTEN  
tcp6 0 0 :::111 :::*  
LISTEN  
tcp6 0 0 :::80 :::*  
LISTEN  
tcp6 0 0 :::25 :::*  
LISTEN  
tcp6 0 0 :::22 :::*  
LISTEN  
tcp6 0 0 :::10050 :::*  
LISTEN  
udp 0 0 127.0.0.54:53 0.0.0.0:*  
udp 0 0 127.0.0.53:53 0.0.0.0:*  
udp 0 0 192.168.1.115:68 0.0.0.0:*  
udp 0 0 0.0.0.0:111 0.0.0.0:*  
udp 0 0 127.0.0.1:161 0.0.0.0:*  
udp6 0 0 :::111 :::*  
udp6 0 0 ::1:161 :::*  
udp6 0 0 fe80::a00:27ff:fe97:546 :::*  
clif@ubuntuuserver:~$
```

Vérification des services actifs

Avec «systemctl list-units --type=service »

```
clif@ubuntuserver: ~$ systemctl list-units --type=service
UNIT
apache-htcacheclean.service
apache2.service
apparmor.service
apport.service
apt-daily-upgrade.service
apt-daily.service
blk-availability.service
console-setup.service
cron.service
dbus.service
finalrd.service
getty@tty1.service
keyboard-setup.service
kmod-static-nodes.service
lvm2-monitor.service
ModemManager.service
multipathd.service
nagios4.service
plymouth-quit-wait.service
plymouth-quit.service
plymouth-read-write.service
polkit.service
postfix.service
postfix@-.service
rpcbind.service
rsyslog.service
setvtrgb.service
snapd.apparmor.service
snapd.seeded.service
```

UNIT	LOAD	ACTIVE	SUB	JOB	DESCRIPTION
apache-htcacheclean.service	loaded	active	running		Disk Cache
apache2.service	loaded	active	running		The Apache
apparmor.service	loaded	active	exited		Load Appa
apport.service	loaded	active	exited		automati
apt-daily-upgrade.service	loaded	activating	start	start	Daily ap
apt-daily.service	loaded	activating	start	start	Daily ap
blk-availability.service	loaded	active	exited		Availabi
console-setup.service	loaded	active	exited		Set cons
cron.service	loaded	active	running		Regular
dbus.service	loaded	active	running		D-Bus Sy
finalrd.service	loaded	active	exited		Create f
getty@tty1.service	loaded	active	running		Getty on
keyboard-setup.service	loaded	active	exited		Set the
kmmod-static-nodes.service	loaded	active	exited		Create L
lvm2-monitor.service	loaded	active	exited		Monitori
ModemManager.service	loaded	active	running		Modem Ma
multipathd.service	loaded	active	running		Device-M
nagios4.service	loaded	active	running		nagios4
plymouth-quit-wait.service	loaded	active	exited		Hold unt
plymouth-quit.service	loaded	active	exited		Terminat
plymouth-read-write.service	loaded	active	exited		Tell Ply
polkit.service	loaded	active	running		Authoriz
postfix.service	loaded	active	exited		Postfix
postfix@-.service	loaded	active	running		Postfix
rpcbind.service	loaded	active	running		RPC bind
rsyslog.service	loaded	active	running		System L
setvtrgb.service	loaded	active	exited		Set cons
snapped.apparmor.service	loaded	active	exited		Load App
snapped.seeded.service	loaded	active	exited		Wait unt
snmpd.service	loaded	active	running		Simple N
ssh.service	loaded	active	running		OpenBSD

lines 1-32

2. Firewall simple avec UFW

Installation de UFW

Par cette commande « `sudo apt install ufw -y` »

```
clif@ubuntuuserver:~$ sudo apt install ufw -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ufw est déjà la version la plus récente (0.36.2-6).
ufw passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 56 non mis à jour.
clif@ubuntuuserver:~$
```

Vérifier l'état du firewall

```
clif@ubuntuuserver:~$ sudo ufw status
Status: inactive
clif@ubuntuuserver:~$
```


Politique par défaut

Bloquer tout :

Avec :

sudo ufw default deny incoming

sudo ufw default allow outgoing

```
clif@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
clif@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
clif@ubuntu:~$ |
```

Autoriser les services nécessaires

#SSH :

sudo ufw allow 22

HTTP :

sudo ufw allow 80

HTTPS :

sudo ufw allow 443

```
clif@ubuntu:~$ sudo ufw allow 22
Rules updated
Rules updated (v6)
clif@ubuntu:~$ sudo ufw allow 80
Rules updated
Rules updated (v6)
clif@ubuntu:~$ sudo ufw allow 443
Rules updated
Rules updated (v6)
clif@ubuntu:~$ |
```


Activer le firewall

```
sudo ufw enable
```

Vérifier :

```
sudo ufw status verbose
```

```
clif@ubuntuserver:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
clif@ubuntuserver:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
22	ALLOW IN	Anywhere
80	ALLOW IN	Anywhere
443	ALLOW IN	Anywhere
22 (v6)	ALLOW IN	Anywhere (v6)
80 (v6)	ALLOW IN	Anywhere (v6)
443 (v6)	ALLOW IN	Anywhere (v6)

```
clif@ubuntuserver:~$ |
```

3. Configuration avec iptables

Afficher les règles actuelles :

`sudo iptables -L`

```
clif@ubuntu: ~  
clif@ubuntu:~$ sudo iptables -L  
Chain INPUT (policy DROP)  
target      prot opt source                destination  
ufw-before-logging-input  all -- anywhere            anywhere  
ufw-before-input  all -- anywhere            anywhere  
ufw-after-input   all -- anywhere            anywhere  
ufw-after-logging-input  all -- anywhere            anywhere  
ufw-reject-input  all -- anywhere            anywhere  
ufw-track-input   all -- anywhere            anywhere  
  
Chain FORWARD (policy DROP)  
target      prot opt source                destination  
ufw-before-logging-forward  all -- anywhere            anywhere  
ufw-before-forward  all -- anywhere            anywhere  
ufw-after-forward   all -- anywhere            anywhere  
ufw-after-logging-forward  all -- anywhere            anywhere  
ufw-reject-forward  all -- anywhere            anywhere  
ufw-track-forward   all -- anywhere            anywhere  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
ufw-before-logging-output  all -- anywhere            anywhere  
ufw-before-output  all -- anywhere            anywhere  
ufw-after-output   all -- anywhere            anywhere  
ufw-after-logging-output  all -- anywhere            anywhere  
ufw-reject-output  all -- anywhere            anywhere  
ufw-track-output   all -- anywhere            anywhere  
  
Chain ufw-after-forward (1 references)  
target      prot opt source                destination  
  
Chain ufw-after-input (1 references)
```

```
clif@ubuntuuserver: ~  
ufw-track-output all -- anywhere anywhere  
  
Chain ufw-after-forward (1 references)  
target prot opt source destination  
  
Chain ufw-after-input (1 references)  
target prot opt source destination  
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:netbios-ns  
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:netbios-dgm  
ufw-skip-to-policy-input tcp -- anywhere anywhere tcp dpt:netbios-ssn  
ufw-skip-to-policy-input tcp -- anywhere anywhere tcp dpt:microsoft-ds  
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:bootps  
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:bootpc  
ufw-skip-to-policy-input all -- anywhere anywhere ADDRTYPE match dst-type BROADCAST  
  
Chain ufw-after-logging-forward (1 references)  
target prot opt source destination limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "  
LOG all -- anywhere anywhere  
  
Chain ufw-after-logging-input (1 references)  
target prot opt source destination limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "  
LOG all -- anywhere anywhere  
  
Chain ufw-after-logging-output (1 references)  
target prot opt source destination  
  
Chain ufw-after-output (1 references)  
target prot opt source destination  
  
Chain ufw-before-forward (1 references)  
target prot opt source destination ctstate RELATED,ESTABLISHED  
ACCEPT all -- anywhere anywhere icmp destination-unreachable  
ACCEPT icmp -- anywhere anywhere  
  
Chain ufw-user-input (1 references)  
target prot opt source destination tcp dpt:ssh  
ACCEPT tcp -- anywhere anywhere udp dpt:22  
ACCEPT udp -- anywhere anywhere tcp dpt:http  
ACCEPT tcp -- anywhere anywhere udp dpt:80  
ACCEPT udp -- anywhere anywhere tcp dpt:https  
ACCEPT tcp -- anywhere anywhere udp dpt:https  
  
Chain ufw-user-limit (0 references)  
target prot opt source destination limit: avg 3/min burst 5 LOG level warn prefix "[UFW LIMIT BLOCK] "  
LOG all -- anywhere anywhere reject-with icmp-port-unreachable  
REJECT all -- anywhere anywhere  
  
Chain ufw-user-limit-accept (0 references)  
target prot opt source destination  
ACCEPT all -- anywhere anywhere  
  
Chain ufw-user-logging-forward (0 references)  
target prot opt source destination  
  
Chain ufw-user-logging-input (0 references)  
target prot opt source destination  
  
Chain ufw-user-logging-output (0 references)  
target prot opt source destination  
  
Chain ufw-user-output (1 references)  
target prot opt source destination  
clif@ubuntuuserver:~$ |
```

Bloquer tout le trafic entrant
sudo iptables -P INPUT DROP

```
clif@ubuntuuserver:~$ sudo iptables -P INPUT DROP  
clif@ubuntuuserver:~$ |
```

Autoriser SSH

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

Autoriser HTTP

```
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

Autoriser HTTPS

```
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

Autoriser les connexions déjà établies

```
sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

```
clif@ubuntuuserver:~$ sudo iptables -A INPUT -p tcp --dport 22 -j
ACCEPT
[sudo] password for clif:
clif@ubuntuuserver:~$ sudo iptables -A INPUT -p tcp --dport 80 -j
ACCEPT
clif@ubuntuuserver:~$ sudo iptables -A INPUT -p tcp --dport 443 -
j ACCEPT
clif@ubuntuuserver:~$ sudo iptables -A INPUT -m conntrack --ctsta
te ESTABLISHED,RELATED -j ACCEPT
clif@ubuntuuserver:~$ |
```

4. Firewall moderne avec nftables

Installer :

```
sudo apt install nftables -y
```

```
clif@ubuntuuserver: ~
x + v
clif@ubuntuuserver:~$ sudo apt install nftables -y
[sudo] password for clif:
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  libnftables1
Paquets suggérés :
  firewallld
Les paquets suivants seront mis à jour :
  libnftables1 nftables
2 mis à jour, 0 nouvellement installés, 0 à enlever et 54 non mis à jour.
Il est nécessaire de prendre 428 ko dans les archives.
Après cette opération, 0 o d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nftables amd64 1.0.9-1ubuntu0.1 [69,8 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnftables1 amd64 1.0.9-1ubuntu0.1 [359 kB]
428 ko réceptionnés en 7s (61,2 ko/s)
(Lecture de la base de données... 95264 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../nftables_1.0.9-1ubuntu0.1_amd64.deb ...
Dépaquetage de nftables (1.0.9-1ubuntu0.1) sur (1.0.9-1build1) ...
Préparation du dépaquetage de .../libnftables1_1.0.9-1ubuntu0.1_amd64.deb ...
Dépaquetage de libnftables1:amd64 (1.0.9-1ubuntu0.1) sur (1.0.9-1build1) ...
Paramétrage de libnftables1:amd64 (1.0.9-1ubuntu0.1) ...
Paramétrage de nftables (1.0.9-1ubuntu0.1) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

Activer :

sudo systemctl enable nftables

```
clif@ubuntuuserver:~$ sudo systemctl enable nftables
Created symlink /etc/systemd/system/sysinit.target.wants/nftables.service → /usr/lib/systemd/system/nftables.service.
clif@ubuntuuserver:~$
```

Exemple de configuration

Éditer :

sudo nano /etc/nftables.conf

clif@ubuntuuser: ~



GNU nano 7.2

/etc/nftables.conf *

```
table inet filter {
chain input {
type filter hook input priority 0;
policy drop;
ct state established,related accept
tcp dport 22 accept
tcp dport 80 accept
tcp dport 443 accept
}
}sudo systemctl start nginx#!/usr/sbin/nft -f

flush ruleset

table inet filter {
    chain input {
        type filter hook input priority filter;
    }
    chain forward {
        type filter hook forward priority filter;
    }
    chain output {
        type filter hook output priority filter;
    }
}
```

^G Help
^X Exit

^O Write Out
^R Read File

^W Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

Appliquer :

sudo systemctl restart nftables

```
clif@ubuntuuserver:~$ sudo systemctl restart nftables
clif@ubuntuuserver:~$ sudo systemctl status nftables
● nftables.service - nftables
   Loaded: loaded (/usr/lib/systemd/system/nftables.service; >
   Active: active (exited) since Mon 2026-03-23 19:37:36 UTC; >
   Docs: man:nft(8)
         http://wiki.nftables.org
   Process: 26813 ExecStart=/usr/sbin/nft -f /etc/nftables.conf >
   Main PID: 26813 (code=exited, status=0/SUCCESS)
   CPU: 13ms

mars 23 19:37:36 ubuntuuserver systemd[1]: Starting nftables.serv>
mars 23 19:37:36 ubuntuuserver systemd[1]: Finished nftables.serv>
lines 1-11/11 (END)
```

5. Tests de sécurité

Tester les ports ouverts :

nmap localhost

```
clif@ubuntuuserver:~$ nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-23 19:46 U
TC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http
111/tcp   open  rpcbind

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
clif@ubuntuuserver:~$ |
```


Durcissement d'un serveur OpenSSH

6. Analyse de la configuration actuelle

Afficher la configuration SSH :

```
systemctl status ssh
```

```
sudo cat /etc/ssh/sshd_config
```

```
clif@ubuntuuser:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-03-18 22:20:39 UTC; 1min 1s ago
     TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
            man:sshd_config(5)
    Main PID: 777 (sshd)
      Tasks: 1 (limit: 4603)
     Memory: 4.1M (peak: 7.9M)
        CPU: 1.287s
     CGroup: /system.slice/ssh.service
            └─777 "sshd: /usr/sbin/sshd -D [listener] 0 of 10->

mars 23 18:58:20 ubuntuuser sshd[26287]: Connection closed by>
mars 23 19:03:20 ubuntuuser sshd[26307]: Connection closed by>
mars 23 19:08:20 ubuntuuser sshd[26328]: Connection closed by>
mars 23 19:13:20 ubuntuuser sshd[26401]: Connection closed by>
mars 23 19:23:20 ubuntuuser sshd[26443]: Connection closed by>
mars 23 19:28:20 ubuntuuser sshd[26665]: Connection closed by>
mars 23 19:33:20 ubuntuuser sshd[26757]: Connection closed by>
mars 23 19:38:20 ubuntuuser sshd[26823]: Connection closed by>
mars 23 19:43:20 ubuntuuser sshd[27295]: Connection closed by>
mars 23 19:48:20 ubuntuuser sshd[27455]: Connection closed by>
lines 1-23/23 (END)
```

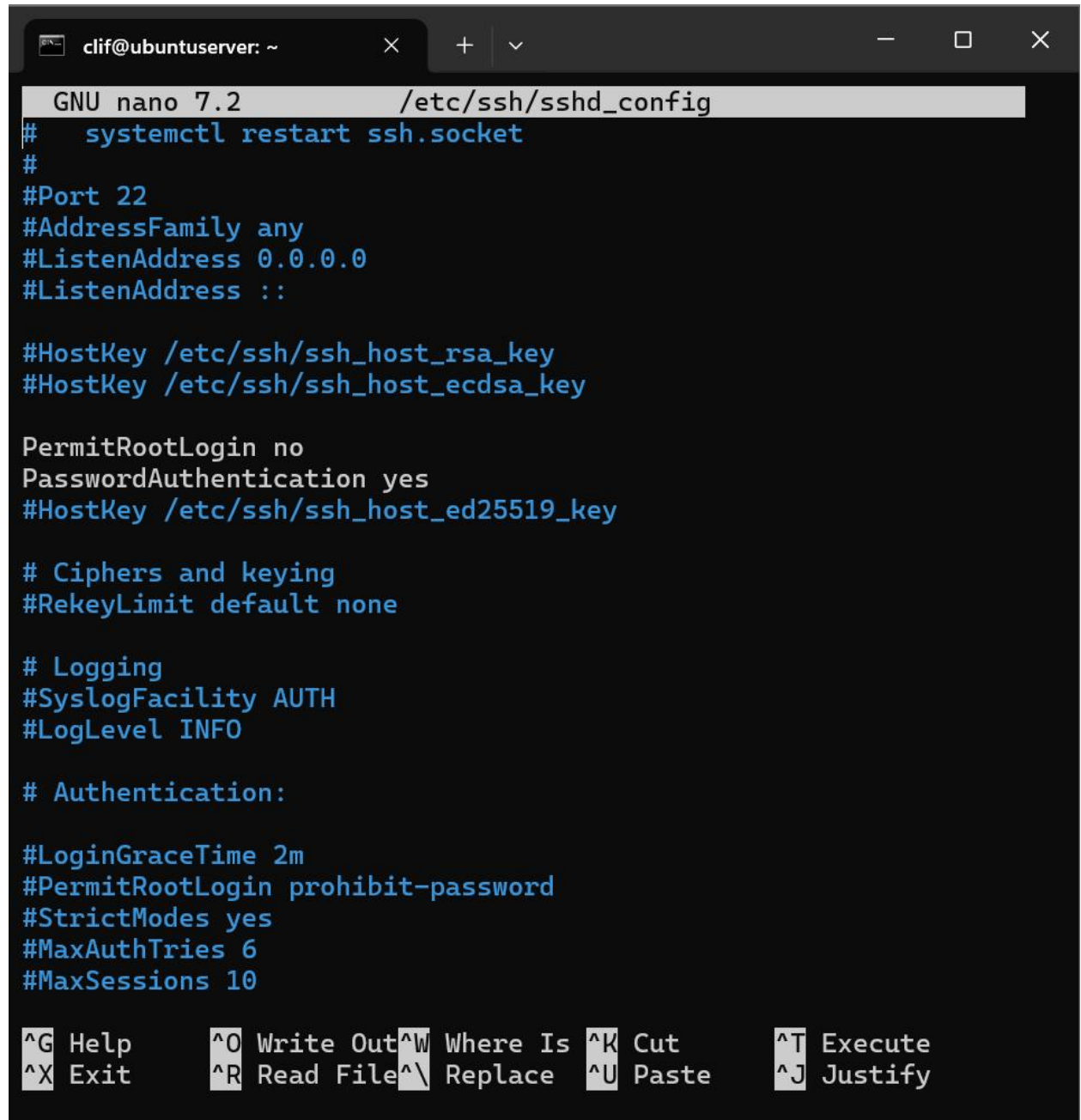
```
clif@ubuntu: ~  
clif@ubuntu:~$ sudo cat /etc/ssh/sshd_config  
  
# This is the sshd server system-wide configuration file. See  
# sshd_config(5) for more information.  
  
# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games  
  
# The strategy used for options in the default sshd_config shipped with  
# OpenSSH is to specify options with their default value where  
# possible, but leave them commented. Uncommented options override the  
# default value.  
  
Include /etc/ssh/sshd_config.d/*.conf  
  
# When systemd socket activation is used (the default), the socket  
# configuration must be re-generated after changing Port, AddressFamily, or  
# ListenAddress.  
#  
# For changes to take effect, run:  
#  
#   systemctl daemon-reload  
#   systemctl restart ssh.socket  
#  
#Port 22  
#AddressFamily any  
#ListenAddress 0.0.0.0  
#ListenAddress ::  
  
#HostKey /etc/ssh/ssh_host_rsa_key  
#HostKey /etc/ssh/ssh_host_ecdsa_key  
  
PermitRootLogin no  
PasswordAuthentication yes
```

```
clif@ubuntuserver: ~  
#X11UseLocalhost yes  
#PermitTTY yes  
PrintMotd no  
#PrintLastLog yes  
#TCPKeepAlive yes  
#PermitUserEnvironment no  
#Compression delayed  
#ClientAliveInterval 0  
#ClientAliveCountMax 3  
#UseDNS no  
#PidFile /run/sshd.pid  
#MaxStartups 10:30:100  
#PermitTunnel no  
#ChrootDirectory none  
#VersionAddendum none  
  
# no default banner path  
#Banner none  
  
# Allow client to pass locale environment variables  
AcceptEnv LANG LC_*  
  
# override default of no subsystems  
Subsystem      sftp      /usr/lib/openssh/sftp-server  
  
# Example of overriding settings on a per-user basis  
#Match User anoncvs  
#      X11Forwarding no  
#      AllowTcpForwarding no  
#      PermitTTY no  
#      ForceCommand cvs server  
clif@ubuntuserver:~$ |
```

7. Désactiver la connexion root

Modifier le fichier :

`sudo nano /etc/ssh/sshd_config`



```
GNU nano 7.2 /etc/ssh/sshd_config
# systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key

PermitRootLogin no
PasswordAuthentication yes
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute
^X Exit      ^R Read File ^\ Replace  ^U Paste    ^J Justify
```

Redémarrer le service :

`sudo systemctl restart ssh`

```
clif@ubuntuuserver:~$ sudo systemctl restart ssh
clif@ubuntuuserver:~$ |
```

8. Authentification par clé SSH

Sur la machine cliente :

Créer une clé :

`ssh-keygen`

```
clif@ubuntuserver:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/clif/.ssh/id_ed25519)
: ssh-copy-id clif@192.168.1.115
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match. Try again.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in ssh-copy-id clif@192.168.1.115
Your public key has been saved in ssh-copy-id clif@192.168.1.115
The key fingerprint is:
SHA256:+LWA5e+VagynJ8ctKYHkgkHKJlu0LTxx7w1gbxcpPds clif@ubuntuse
rver
The key's randomart image is:
+--[ED25519 256]--+
|
| .o + . =
|+o * + ..=
|+o* . ==o E
|o= o oo+S .
|o . o ..o+.. .
|   + . ..*ooo
|   . . +.Oo.
|           *o.
+-----[SHA256]-----+
clif@ubuntuserver:~$ |
```

Tester la connexion :

```
clif@ubuntu: ~  
clif@ubuntu:~$ ssh clif@192.168.1.115  
The authenticity of host '192.168.1.115 (192.168.1.115)' can't be established.  
ED25519 key fingerprint is SHA256:3VZE/uk3h8H/44ROT7FI6rJgr6e6Qq2QGsu0JRVXJwo.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.1.115' (ED25519) to the list of known hosts.  
clif@192.168.1.115's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of lun. 23 mars 2026 20:04:47 UTC  
  
System load:          0.0  
Usage of /:           50.5% of 11.21GB  
Memory usage:         9%  
Swap usage:           0%  
Processes:            210  
Users logged in:      1  
IPv4 address for enp0s3: 192.168.1.115  
IPv6 address for enp0s3: fdcd:7bb5:f446:8:a00:27ff:fe97:c322  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
  https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
La maintenance de sécurité étendue pour Applications n'est pas activée.  
  
61 mises à jour peuvent être appliquées immédiatement.  
53 de ces mises à jour sont des mises à jour de sécurité.
```



```
clif@ubuntuserver: ~  
* Management:      https://landscape.canonical.com  
* Support:         https://ubuntu.com/pro  
  
System information as of lun. 23 mars 2026 20:04:47 UTC  
  
System load:                0.0  
Usage of /:                 50.5% of 11.21GB  
Memory usage:               9%  
Swap usage:                 0%  
Processes:                  210  
Users logged in:            1  
IPv4 address for enp0s3:    192.168.1.115  
IPv6 address for enp0s3:    fdcd8:7bb5:f446:8:a00:27ff:fe97:c322  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
  https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
La maintenance de sécurité étendue pour Applications n'est pas activée.  
  
61 mises à jour peuvent être appliquées immédiatement.  
53 de ces mises à jour sont des mises à jour de sécurité.  
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable  
  
1 mise à jour de sécurité supplémentaire peut être appliquée avec ESM Apps  
En savoir plus sur l'activation du service ESM Apps at https://ubuntu.com/esm  
  
Last login: Mon Mar 23 17:22:11 2026 from 192.168.1.220  
clif@ubuntuserver:~$ |
```

9. Désactiver l'authentification par mot de passe

Modifier :

PasswordAuthentication no


```
clif@ubuntuuserver: ~
GNU nano 7.2 /etc/ssh/sshd_config *
# When systemd socket activation is used (the default), the soc>
# configuration must be re-generated after changing Port, Addre>
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key

PermitRootLogin no
PasswordAuthentication no
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

sudo systemctl restart ssh

```
clif@ubuntuuserver:~$ sudo systemctl restart ssh
[sudo] password for clif:
clif@ubuntuuserver:~$ |
```

10. Changer le port SSH

Modifier :

Port 2222

```
clif@ubuntuserver: ~  
GNU nano 7.2 /etc/ssh/sshd_config *  
# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/b>  
  
# The strategy used for options in the default sshd_config ship>  
# OpenSSH is to specify options with their default value where  
# possible, but leave them commented. Uncommented options over>  
# default value.  
  
Include /etc/ssh/sshd_config.d/*.conf  
  
# When systemd socket activation is used (the default), the soc>  
# configuration must be re-generated after changing Port, Addre>  
# ListenAddress.  
#  
# For changes to take effect, run:  
#  
#   systemctl daemon-reload  
#   systemctl restart ssh.socket  
#  
#Port 2222  
#AddressFamily any  
#ListenAddress 0.0.0.0  
#ListenAddress ::  
  
#HostKey /etc/ssh/ssh_host_rsa_key  
#HostKey /etc/ssh/ssh_host_ecdsa_key  
  
PermitRootLogin no  
PasswordAuthentication no  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

sudo systemctl restart ssh

```
clif@ubuntuserver:~$ sudo systemctl restart ssh  
clif@ubuntuserver:~$ |
```

Connexion :

```
clif@ubuntuuserver:~$ sudo nano /etc/ssh/sshd_config
clif@ubuntuuserver:~$ ssh -p 2222 clif@192.168.1.115
ssh: connect to host 192.168.1.115 port 2222: Connection refused
clif@ubuntuuserver:~$ sudo ufw allow 2222/tcp
Rule added
Rule added (v6)
clif@ubuntuuserver:~$ sudo ufw reload
Firewall reloaded
clif@ubuntuuserver:~$ ssh -p 2222 clif@192.168.1.115
ssh: connect to host 192.168.1.115 port 2222: Connection refused
clif@ubuntuuserver:~$ sudo nano /etc/ssh/sshd_config
clif@ubuntuuserver:~$ ^C
clif@ubuntuuserver:~$ sudo systemctl daemon-reload
clif@ubuntuuserver:~$ sudo systemctl restart ssh
clif@ubuntuuserver:~$ sudo ss -tlnp | grep sshd
LISTEN 0      128          0.0.0.0:2222      0.0.0.0:*        users
:(("sshd",pid=28569,fd=3))

LISTEN 0      128          [::]:2222        [::]:*          users
:(("sshd",pid=28569,fd=4))

clif@ubuntuuserver:~$ ssh -p 2222 clif@192.168.1.115
clif@192.168.1.115: Permission denied (publickey).
clif@ubuntuuserver:~$ |
```

11. Protection contre brute force

Installation :

```
sudo apt install fail2ban -y
```

```
clif@ubuntuuserver: ~$ sudo apt install fail2ban -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  python3-pyasyncore python3-pyinotify whois
Paquets suggérés :
  monit sqlite3 python-pyinotify-doc
Les NOUVEAUX paquets suivants seront installés :
  fail2ban python3-pyasyncore python3-pyinotify whois
0 mis à jour, 4 nouvellement installés, 0 à enlever et 54 non mis à jour.
Il est nécessaire de prendre 496 ko dans les archives.
Après cette opération, 2 572 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyasyncore all 1.0.2-2 [10,1 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 fail2ban all 1.0.2-3ubuntu0.1 [409 kB]
Réception de :3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyinotify all 0.9.6-2ubuntu1 [25,0 kB]
Réception de :4 http://archive.ubuntu.com/ubuntu noble/main amd64 whois amd64 5.5.22 [51,7 kB]
496 ko réceptionnés en 28s (17,9 ko/s)
Sélection du paquet python3-pyasyncore précédemment désélectionné.
(Lecture de la base de données... 96135 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../python3-pyasyncore_1.0.2-2_all.deb ...
Dépaquetage de python3-pyasyncore (1.0.2-2) ...
Sélection du paquet fail2ban précédemment désélectionné.
Préparation du dépaquetage de .../fail2ban_1.0.2-3ubuntu0.1_all.deb ...
Dépaquetage de fail2ban (1.0.2-3ubuntu0.1) ...
Sélection du paquet python3-pyinotify précédemment désélectionné.
Préparation du dépaquetage de .../python3-pyinotify_0.9.6-2ubuntu1_all.deb ...
Dépaquetage de python3-pyinotify (0.9.6-2ubuntu1) ...
Sélection du paquet whois précédemment désélectionné.
Préparation du dépaquetage de .../whois_5.5.22_amd64.deb ...
Dépaquetage de whois (5.5.22) ...
Paramétrage de whois (5.5.22) ...
Paramétrage de python3-pyasyncore (1.0.2-2) ...

/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:451: SyntaxWarning: invalid escape sequence '\s'
_test_variants('common', prefix="\s*\$+ sshd\[<F-MLFID>\d+</F-MLFID>[:]\s+")
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:537: SyntaxWarning: invalid escape sequence '\['
'common[prefregex="^svc\[<F-MLFID>\d+</F-MLFID>\] connect <F-CONTENT>.+</F-CONTENT>$"'
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1375: SyntaxWarning: invalid escape sequence '\s'
    '{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-mp\s+.*\s+\\Khandle\s+(\d+)\$'; } | while read -r hdl; d
o' ",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1378: SyntaxWarning: invalid escape sequence '\s'
    '{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-mp\s+.*\s+\\Khandle\s+(\d+)\$'; } | while read -r hdl;
do' ",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1421: SyntaxWarning: invalid escape sequence '\s'
    '{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-nft-ap\s+.*\s+\\Khandle\s+(\d+)\$'; } | while read -r hdl; d
o' ",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1424: SyntaxWarning: invalid escape sequence '\s'
    '{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-ap\s+.*\s+\\Khandle\s+(\d+)\$'; } | while read -r hdl;
do' ",
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
Paramétrage de python3-pyinotify (0.9.6-2ubuntu1) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on
this host.
clif@ubuntuuserver:~$ |
```

Vérifier :

sudo systemctl status fail2ban


```
clif@ubuntuuserver:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; >
   Active: active (running) since Mon 2026-03-23 20:33:44 UTC >
     Docs: man:fail2ban(1)
   Main PID: 28804 (fail2ban-server)
     Tasks: 5 (limit: 4603)
    Memory: 25.2M (peak: 25.5M)
       CPU: 573ms
    CGroup: /system.slice/fail2ban.service
            └─28804 /usr/bin/python3 /usr/bin/fail2ban-server >

mars 23 20:33:44 ubuntuuserver systemd[1]: Started fail2ban.serv>
mars 23 20:33:44 ubuntuuserver fail2ban-server[28804]: 2026-03-2>
mars 23 20:33:45 ubuntuuserver fail2ban-server[28804]: Server re>
lines 1-14/14 (END)
```

Afficher les jails actives :

```
sudo fail2ban-client status
```

```
clif@ubuntuuserver:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:  sshd
clif@ubuntuuserver:~$
```

12. Tests de sécurité

Vérifier les ports ouverts :

ss -tuln

```

clif@ubuntu:~$ ss -tuln
Netid State  Recv-Q Send-Q           Local Address:Port           Peer Address:Port           Process
udp    UNCONN  0      0           127.0.0.54:53                  0.0.0.0:*
udp    UNCONN  0      0           127.0.0.53%lo:53              0.0.0.0:*
udp    UNCONN  0      0       192.168.1.115%enp0s3:68        0.0.0.0:*
udp    UNCONN  0      0           0.0.0.0:111                   0.0.0.0:*
udp    UNCONN  0      0           127.0.0.1:161                 0.0.0.0:*
udp    UNCONN  0      0           [::]:111                      [::]:*
udp    UNCONN  0      0           [::1]:161                     [::]:*
udp    UNCONN  0      0       [fe80::a00:27ff:fe97:c322]%enp0s3:546 [::]:*
tcp    LISTEN  0      4096           0.0.0.0:111                   0.0.0.0:*
tcp    LISTEN  0      100           0.0.0.0:25                    0.0.0.0:*
tcp    LISTEN  0      128           0.0.0.0:2222                  0.0.0.0:*
tcp    LISTEN  0      4096           127.0.0.53%lo:53              0.0.0.0:*
tcp    LISTEN  0      4096           127.0.0.54:53                 0.0.0.0:*
tcp    LISTEN  0      4096           0.0.0.0:10050                 0.0.0.0:*
tcp    LISTEN  0      4096           [::]:111                      [::]:*
tcp    LISTEN  0      511           *:80                          *:
tcp    LISTEN  0      100           [::]:25                       [::]:*
tcp    LISTEN  0      128           [::]:2222                     [::]:*
tcp    LISTEN  0      4096           [::]:10050                    [::]:*
clif@ubuntu:~$

```

nmap IP_SERVEUR

```
clif@ubuntuserver:~$ nmap 192.168.1.115
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-23 20:41 UTC
Nmap scan report for ubuntuserver (192.168.1.115)
Host is up (0.00044s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
25/tcp    open  smtp
80/tcp    open  http
111/tcp   open  rpcbind
2222/tcp  open  EtherNetIP-1

Nmap done: 1 IP address (1 host up) scanned in 5.30 seconds
clif@ubuntuserver:~$ |
```

Réponses aux Questions ?

1. Différence entre firewall stateful et stateless

- **Stateless** : applique des règles simples sans tenir compte du contexte. Chaque paquet est filtré isolément (ex. : “bloque tout ce qui vient du port 80”).
- **Stateful** : garde en mémoire l’état des connexions (table de suivi). Il sait si un paquet appartient à une session déjà autorisée. → Plus intelligent et sécurisé.

2. Avantages de nftables par rapport à iptables

- Syntaxe plus simple et cohérente.
- Unifie IPv4, IPv6 et ARP dans une seule interface.
- Meilleure performance (un seul moteur).
- Plus flexible : permet des règles dynamiques et des scripts plus propres.

3. Pourquoi toujours autoriser SSH avant d’activer un firewall

- Si tu bloques tout sans exception, tu risques de **te couper l’accès à ton serveur**.
- Autoriser SSH en premier garantit que tu gardes une porte d’entrée pour administrer et corriger la configuration.

4. Pourquoi désactiver la connexion root

- Le compte root est une cible privilégiée des attaques.
- Si root est désactivé, un attaquant doit deviner **à la fois l’utilisateur et le mot de passe/clé**.
- On utilise un compte normal + sudo → sécurité renforcée.

5. Avantages de changer le port SSH

- Réduit les attaques automatiques (bots scannent surtout le port 22).
- Diminue le bruit dans les logs.
- Ce n'est pas une sécurité absolue, mais une **mesure d'obscurcissement** utile en complément.

6. Rôle de Fail2Ban

- Surveille les logs (ex. SSH, Apache).
- Détecte les tentatives répétées d'attaque (bruteforce, scans).
- Ajoute automatiquement une règle firewall pour **bannir l'IP fautive** pendant un certain temps.

En résumé :

- **Stateful > Stateless** pour la sécurité.
- **Nftables** est moderne et plus performant qu'iptables.
- Toujours autoriser **SSH** avant firewall.
- **Root désactivé** = moins de risques.
- **Changer le port SSH** = moins d'attaques automatiques.
- **Fail2Ban** = protection active contre les bruteforce.